

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a Delta table named "customer" with schema: CustId, Name, EmailId, Region
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a Delta table named "order" with schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId
3. Add a new column "TotalAmount" to the "order" table, calculated as the product of PricePerUnit and Qty

FR-DLT-002: Clean and Transform Customer and Order Data

Title: Clean and Transform Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables are loaded with data

Main Flow / Functional Steps:

1. Remove null records from the "customer" and "order" Delta tables
2. Remove duplicate records from the "customer" and "order" Delta tables

FR-DLT-003: Create and Load Order Summary Table

Title: Create and Load Order Summary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined customer and order data using SCD Type 2.

Preconditions:

- Customer and order Delta tables are cleaned and transformed
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table in catalog "gen_ai_poc_databrickscoe" and schema "sdhc_wizard" if it does not exist
2. Join the "customer" and "order" Delta tables on CustId
3. Load the joined data into the "ordersummary" table using SCD Type 2, with schema: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date

FR-DLT-004: Update Order Summary Table on Customer Changes

Title: Update Order Summary Table on Customer Changes

Description: The system must update the "ordersummary" table when there are changes to the customer data.

Preconditions:

- Customer data is updated
- "ordersummary" table exists and is loaded with data

Main Flow / Functional Steps:

1. Identify changes to the customer data
2. Update the "ordersummary" table by making old records inactive and new records active, with corresponding changes to StartDate and EndDate

FR-DLT-005: Create and Load Customer Aggregate Spend Table

Title: Create and Load Customer Aggregate Spend Table

Description: The system must create a "customeraggregatespend" table and load it with aggregated data from the "ordersummary" table.

Preconditions:

- "ordersummary" table exists and is loaded with data

- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table in catalog "gen_ai_poc_databrickscoe" and schema "sdhc_wizard" if it does not exist, with schema: Name, TotalAmount, Date
2. Aggregate the TotalAmount column from the "ordersummary" table, grouped by Name and Date
3. Load the aggregated data into the "customeraggregatespend" table

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement Delta Live Tables (DLT) to execute the above steps.

Preconditions:

- All previous steps are implemented and tested

Main Flow / Functional Steps:

1. Implement DLT to execute steps 1 to 5 above